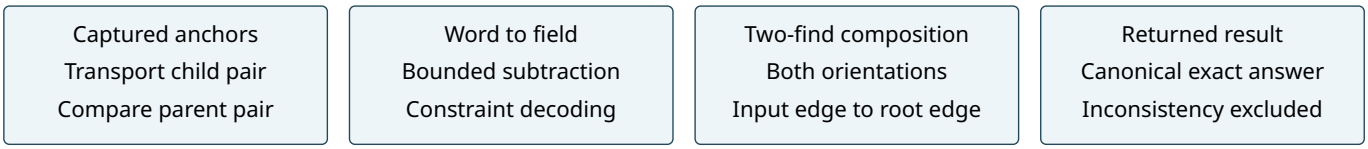
Anchored union return: source order, decoding and consistency

Producer T-1340; archive T-1343. This packet models the distinct-root return path of SignedGraph::edge at 336c98381b. It captures child and parent anchors before graph updates, transports the child anchor, and compares it with an existing parent anchor. It retains an explicit inconsistent result to model the assertion branch.



Result. Ten named declarations compile. The word result decodes to the corresponding field constraint. Under the signed incoming relation and existing graph/anchor invariant, that constraint is safe: it is absent, or its answer equals the target parameter; the inconsistent branch is excluded. A returned word is canonical, and field equality yields exact equality to a canonical target word.

Composition. Safety holds in both union orientations, including the reversed potential, and after the two word-level find calls. The proof derives the reduced root equation from the original signed edge. A source-order lemma shows that saved pre-update anchors followed by the anchor call on the updated state produce the same modeled return. These results complement the earlier union state-preservation and size proofs.

Checks. Seven concrete word-level controls at modulus seven pass: neither anchor; child only; parent only; equal anchors; inconsistent anchors; and forward/reverse answers, each equal to six for the chosen consistent coefficients. Seventeen separate named axiom readbacks contain only propext, Classical.choice and Quot.sound. Empty, renamed and sorryAx inventories are rejected. A false claim that conflicting equal-slope anchors return no constraint is rejected by Lean.

Assumptions. The decoding and safety theorems use a prime UInt64 modulus with $1 < r$ and $r <= 2^{63}$, canonical coefficients and anchors, signs in +1,-1, a valid input graph and a true incoming signed relation. Find composition additionally assumes the forest invariant and the stated word runs. Existing root values and incoming relation truth remain premises, not newly established geometric facts. The general return lemmas are valid for arbitrary supplied indices; the native distinct-root dispatch selects their intended use.

Boundary and evidence. This is a hand-written source-bound model, not a Rust compiler refinement. Full dispatcher composition, counters and allocation semantics remain separate. Independent source-fidelity review is pending. Imported compiled dependencies are reused and hashed; no independent kernel/binary recheck occurred. Pre/post hashes do not exclude transient concurrent edits. Each module check used one thread, 4096 MiB and 120 seconds. The initial decoding/field proof failed and is retained; all later successful attempts and the deliberate false theorem are retained separately. No scientific run, cost measurement, cohort change or budget renewal occurred.

Next construction. Combine same-root and distinct-root results under the source dispatcher with the size-selected orientation, then attach exact counter accounting. The full cohort and independent-review obligations remain in the coverage audit; this packet changes the returned-constraint proof obligation only.